

WHITE PAPER COMPANION · II

SILICONA

Benchmark Protocol

Formal Measurement & Reporting Standard —
v1.0. How a SILICONA benchmark may be run,
repeated, and reported: pinned toolchains,
recorded seeds, distributions instead of
headlines, and negative results published as
data.

SILICONA Engineering

Companion to the Technical White Paper — 12 September 2026

Applies to all SILICONA benchmark releases

EVIDENCE · MEASUREMENT · VERIFICATION

Contents

- 1. Purpose and Claim Standard 1**
- 2. Benchmark Families 2**
 - The L1–L10 certification ladder 2
 - The three search families 2
- 3. Protocol Rules 3**
 - R1 — Repetition and seeds 3
 - R2 — Environment pinning 3
 - R3 — Failure handling 4
 - R4 — Acceptance 5
- 4. Reporting Standard 5**
- 5. Campaign Phases 6**
 - Phase A — Internal repeat campaign (current) 6
 - Phase B — Cross-provider measurement campaign 6
 - Phase C — External anchors 6
- 6. Governance 6**
 - Release authority and signature 6
 - Retraction policy 7
 - Reproducibility 7
 - Document control 7

1. Purpose and Claim Standard

This protocol defines what a SILICONA benchmark is, how it is run, and how it may be reported. It exists because the agentic-EDA market publishes dramatic multipliers with no published methodology — “over 40X faster RTL validation” [Cadence, vendor press release], “up to 50X faster time-to-validated RTL” [Synopsys, vendor press release], “11x faster bug resolution” [Silimate, vendor page] — and none of the fourteen primary sources in the companion competitor-intelligence chapter publishes the protocol, sample size, seeds, variance or environment behind its numbers. SILICONA’s answer is not a counter-multiplier; it is a standard under which its own numbers can be checked, repeated and — where they are wrong — retracted. This document is that standard, applied to every benchmark number SILICONA publishes, starting with the ones already in the engineering-memory database.

A SILICONA benchmark may assert only what a deterministic tool run recorded. The platform enforces three evidence states on every reported figure, and the protocol inherits them as the claim standard:

Evidence state	Meaning	Protocol consequence
MEASURED	The figure is the direct output of a deterministic tool run on a recorded artifact (simulation verdict, formal result, synthesis statistic, timing number, DRC/LVS verdict, search QoR)	Reportable with per-run provenance; may enter headline results only with n, spread and environment disclosed
ANALYTICAL ESTIMATE	The figure is derived from measured inputs by a declared analytical model (pre-route power from measured cell counts; timing before parasitics)	Must carry an explicit ESTIMATE label and a methodology disclosure; never mixed into measured headlines
CONFLICTING EVIDENCE	Two sources of evidence disagree (e.g., a narrated expectation vs. a tool verdict)	The conflict is reported as a conflict; the tool verdict governs acceptance, the conflict is recorded as data

Table 1: The three enforced evidence states — the claim standard for every SILICONA number.

Two boundaries of the claim standard follow from the white paper’s ledger and are binding here. First, the physical claim ceiling: SILICONA claims demonstrated open PDK closure — the sky130A flow reaching DRC 0 violations, an LVS MATCH dual verdict (foundry deck plus deterministic structural compare) and GDS-II streamout — and nothing beyond it; post-route extraction and signoff STA are explicitly unclaimed. When this protocol quotes a competitor’s “sign-off” language, the quotation marks carry the attribution. Second, the standing do-not-publish rules: no iteration-time-versus-human claims, no competitor-superiority claims, no production-readiness claims. The house rule this document operationalizes: AI proposes; deterministic EDA tools measure; verification decides; every result recorded as traceable engineering evidence.

n≥5	0	51	1–7%
repetitions per family per mode	headline percentages without n and spread	published HTTP-429 failures	disclosed run-to-run toolchain variance

2. Benchmark Families

SILICONA benchmarks live on two surfaces: the certification ladder (L1–L10), which measures the platform’s engineering capability end to end, and the search families, which measure the AI layer’s contribution against deterministic baselines under identical toolchains. Both surfaces produce the same currency — tool-verdicted runs recorded in the database — and both are in scope for every rule in Section 3.

The L1–L10 certification ladder

Level	Name	Evidence criterion (as implemented)	Status
L1	Combinational block	Qualifying run with lint PASS + sim PASS and zero sequential cells (yosys stat on the mapped netlist)	CERTIFIED
L2	Stateful block with formal proof	Certified sync-FIFO run with formal PASS (SymbiYosys + Z3)	CERTIFIED
L3	Protocol IP (UART TX)	Project exists with runs (AI path) — template until a qualifying full-pipeline run	TEMPLATE
L4	Optimized compute block (ALU)	Project exists with runs — template until qualifying run	TEMPLATE
L5	Subsystem integration	Qualifying full-pipeline PASS on a multi-block subsystem	CERTIFIED
L6	Processor (RV32I)	Project exists with runs — template until qualifying run	TEMPLATE
L7	Accelerator discovery	A grid SearchRun whose discovered winner passes the full pipeline	CERTIFIED (grid families)
L8	Complete SoC	Qualifying run on the SoC top with the full pipeline	TEMPLATE
L9	Complete physical implementation	Latest revision passes physical design + GDS + DRC 0 + LVS MATCH	CERTIFIED — demonstrated open-PDK closure (rev-22; re-certified with post-route timing at rev-24)
L10	Architectures beating human-selected baselines	A SearchRun (rl/multi) with beatBaseline and discovery recorded	MEASURED — with honest negatives (Section 4)

Table 2: The certification ladder, one line per level, with current status per the evidence pack.

The three search families

Search benchmarks measure whether AI-proposed architecture changes actually improve measured quality-of-results over a human-selected baseline configuration of the same family, under the same objective (area×delay product, ADP) and the same toolchain. Three families are currently in the corpus, each runnable in three modes:

Family (domain)	Search space	Objective and gate	Modes
matrix-mac (accelerator)	Parallel dot-product kernel: 4 lane counts × 3 operand widths (12 configurations)	ADP on measured synthesis + timing; functional gate before any measurement	grid / rl / multi
fir-filter (DSP)	Serial FIR filter: 4 tap counts × 2 sample widths (8 configurations)	ADP on measured synthesis + timing; functional gate before any measurement	grid / rl / multi
stream-fifo (stream)	Fixed 512-bit storage budget decomposed into DEPTH × WIDTH (6 depth choices)	ADP on measured synthesis + timing; functional gate before any measurement	grid / rl / multi

Table 3: Search families and modes. Every candidate — winner, loser, duplicate, reject — is recorded.

- **grid** — full enumeration of the family’s parameter combinations; every candidate measured by the real toolchain; the winner qualified through the full pipeline as a discovered project.
- **rl** — ϵ -greedy Q-learning over the same space; state = configuration, action = move to neighbor, reward = the **measured** ADP improvement of the neighbor from a real tool run; rewards are never narrated.
- **multi** — the model engine arena: five proposing engines (uniform-random control, information-blind LLM, three memory-informed LLM strategies) compete in one search; a router learns from measured per-engine rewards; the whole population is recorded — losers, rejects and duplicates included.

3. Protocol Rules

This section is the core of the standard. A benchmark that violates any rule below is not a SILICONA benchmark and may not be published under the SILICONA name.

R1 — Repetition and seeds

- **n≥5 repetitions** per family per mode before any improvement claim; the current corpus carries n=5–6 per family on the multi-engine arena mode, and Phase A (Section 5) exists to extend repetition coverage.
- **Seeds are recorded** with every run — AI-sampling seeds and search RNG state — so the run is replayable in principle and variance is attributable.
- **Fresh context per AI episode.** Each proposing engine works from a fresh context; no hidden memory of prior runs unless the memory mechanism is itself the declared treatment (the arena’s memory-informed strategies are labeled as such).
- **Winners are re-measured fresh.** A memory-cited winner configuration is re-run through the real toolchain before any claim; citation never substitutes for measurement.

R2 — Environment pinning

Every benchmark release pins the complete deterministic environment to the platform toolchain manifest. The current manifest (deployment as of 12 September 2026):

Layer	Component	Pinned version
RTL and simulation	Icarus Verilog (iverilog + vvp)	iverilog 14.0
Lint / language services	slang	11.0.448
Cross-check simulator	Verilator (present, not in the MVP flow)	5.051
Synthesis	Yosys	0.68+136
Formal	SymbiYosys (sby) + Z3	sby 0.68, z3 4.15.5
Equivalence	eqy (RTL vs mapped netlist hard gate)	0.68
Timing	OpenSTA (sta binary)	2.6.0
Physical implementation	OpenROAD	v2.0-17598
Physical verification	KLayout (GDS streamout, DRC, LVS)	0.30.0
PDK	sky130A (volare / open_pdks)	open_pdks fa87f8f4
Standard-cell library	sky130_fd_sc_hd	tt_025C_1v80 (nom.tlef + 437-cell LEF + liberty)

Table 4: Pinned environment — the platform toolchain manifest (OSS CAD Suite 2026-08-28 deployment plus rootless packages).

A benchmark number published without its manifest row is a protocol violation. When the deployment upgrades a component, previously published numbers keep their original manifest and new numbers carry the new one; versions are never silently rebased.

R3 — Failure handling

- **Provider failures are data.** A model-evaluation call that fails is recorded as a failed row with its HTTP status — the 51 rate-limit (HTTP 429) failures among 177 recorded evaluations are the precedent, published, not smoothed away.
- **Tool failures are data.** A toolchain crash, OOM or timeout becomes a failure row with the stderr captured; it never disappears into a retry loop.
- **A failed family is published as failed.** The matrix-mac/grid runs FAILED (twice) and are published as failures in the results pack; the protocol forbids re-running a failed family until it passes and reporting only the passing attempt.
- **Honest negatives are mandatory, not optional.** The stream-fifo RL result — a 0.00% improvement where the uniform-random control matched the LLM engines on a monotonic objective — is part of the L10 record precisely because it bounds what the claims can mean.

R4 — Acceptance

Only deterministic tool verdicts accept or reject work. AI-proposed artifacts pass gates by passing them: lint at zero errors, simulation to a TB_PASS marker, formal results stated with their bound (bounded model checking depth is part of the claim), equivalence proven by eqy, timing measured by OpenSTA, DRC/LVS verdicts from KLayout on the streamed GDS. Nothing in the protocol accepts narrated quality, and no gate may be widened post-hoc to admit a candidate.

4. Reporting Standard

The reporting rules exist to make every published number re-derivable and every unmeasured gap visible. They apply to benchmark releases, the white paper, investor material and any public claim that cites a SILICONA measurement.

- **Distributions, not headlines.** Any improvement figure is reported as min / median / max (IQR where $n \geq 5$) with the full per-run table — e.g., the fir-filter multi-mode result is reported as five runs averaging +79.65% spanning 74.15–83.32%, not as “80% faster”.
- **No single headline percentage without n and spread.** A bare multiplier or percentage is a protocol violation, whatever its direction or owner.
- **Negative results are mandatory.** Failed families, honest zeros and losing engines appear in every release that covers their family or mode.
- **Per-run provenance is part of the result.** Every row carries: revision id, provider + model tag, seed, tool versions (manifest id), and the verdict chain from gate to gate.
- **Variance is disclosed per claim.** The observed 1–7% run-to-run toolchain variance on repeated identical inputs is disclosed wherever it could affect a comparison; a difference inside that band is reported as “within toolchain variance”, not as a win.
- **Single-run results are labeled single-run.** The L9 physical reference (rev-24: 950 mapped cells, $18,672.9 \mu\text{m}^2$, Fmax 240 MHz, post-route slack +3.48 ns, DRC 0, LVS MATCH dual verdict, 2,361,570-byte GDS) is published as a measured single-design result with its tool attribution, not as a general performance claim.

Family / mode	n	Improvement over baseline	Reported as
matrix-mac / multi	5	+98.21% (± 0.00 across runs)	Distribution: identical five-run set; winner re-measured fresh
fir-filter / multi	5	avg +79.65%, range 74.15–83.32%	Distribution with full per-run table
stream-fifo / multi	6	+34.30%	Distribution; six recorded runs
stream-fifo / rl	1	0.00%	Honest negative: random control matched the LLM engines on a monotonic objective
matrix-mac / grid	2	FAILED	Published as failed; no improvement claimed

Table 5: The current measured search record, as the reporting standard requires it — including the negative and the failure.

5. Campaign Phases

Phase A — Internal repeat campaign (current)

Extend the existing corpus to full protocol coverage on the current families: bring every family $\times$ mode combination to $n \geq 5$ with recorded seeds, re-measure winners fresh, and publish the expanded per-run tables. The existing multi-mode sets ($n=5-6$ per family) are the starting point; the grid and rl modes are the coverage gaps. Phase A runs on the measured GLM primary with the environment of Table 4 and produces the baseline distributions every later phase is compared against.

Phase B — Cross-provider measurement campaign

Repeat the same families, same seeds and same pinned environment across the 12-provider registry — OpenAI, Anthropic, Google, xAI, DeepSeek, Mistral, Alibaba Qwen, Meta, Groq, OpenRouter, the generic OpenAI-compatible slot, and the measured GLM primary — with provider-labeled ledgers. The registry is an architectural claim today; Phase B is what converts it into measured performance, and it starts when provider keys arrive. The published output is per-provider distributions under identical deterministic gates — a cross-model comparison nobody in the landscape currently offers — with the single-provider constraint of the current corpus (all 177 evaluation rows are GLM; the 51 rate-limit failures are the visible dependency) stated as the baseline condition.

Phase C — External anchors

Anchor SILICONA measurements to external benchmarks where they exist. NVIDIA’s CVDP verification benchmark is observed as the only named external AI benchmark in the current competitive landscape (one vendor, Agentrys, claims to be “among the first to exceed 90% accuracy” on it, per its own press release). Participation in CVDP or comparable public benchmarks is a future campaign item under this protocol — with results reported exactly as SILICONA reports its own families: n , seeds, spread, failures. It is not a present claim; the protocol’s contribution here is that when it happens, the methodology will already be binding.

6. Governance

Release authority and signature

Benchmark releases are issued by the owner. A release comprises: the results pack (per-run tables with provenance), the manifest id of the pinned environment, the claim ledger mapping every headline to its rows, and the explicit list of what is not claimed. Material that has not passed the Section 4 reporting standard may not carry the SILICONA name in any channel.

Retraction policy

Retractions are recorded, never silent. When a published number is found wrong — a mismeasured baseline, a gate that should have failed, an environment that was not in fact pinned — the correction is published as a dated retraction referencing the original release, the original release stays on the record, and the affected claim is marked in the claim ledger. The evidence-pack precedent (failures published as failures) extends to corrections: the point of the system is that its history is auditable, which requires the mistakes to remain visible.

Reproducibility

Any published run is replayable in principle and in practice: the pinned toolchain, the PDK and library versions, the provider + model tag and the recorded seed together determine the deterministic tool verdicts. Reproduction of an AI-proposing step is treated as distribution, not identity — models are not deterministic across providers; the deterministic half of the loop (every gate that decides acceptance) must reproduce exactly, and that asymmetry is stated in every release. The strongest claim this protocol licenses is therefore the modest one the market currently lacks: every SILICONA number can be checked, and this document is the standing instructions for checking it.

Document control

Field	Value
Protocol version	v1.0 (first issue)
Issue date	12 September 2026
Status	In force for all SILICONA benchmark releases from this date
Supersedes	None — first formal issue; prior white-paper methodology sections remain consistent with it
Review triggers	Phase B start (cross-provider campaign); any toolchain-manifest change; any retraction event
Companion documents	SILICONA Technical White Paper (v1.1); SILICONA Competitor Intelligence (v1.0)

Table 6: Protocol document control.